



http://elinux.org/RPi_VerifiedPeripherals#Working_USB_Wi-Fi_Adaptors

A Wi-Fi adaptor consumes more power because it transmits energy into the air. A USB hub that has a small hole to receive 5V DC supply will be suitable for your Raspberry Pi.

Now connect USB and additional 5V supply to the Raspberry Pi. Connect the USB Wi-Fi adaptor using any of the port of the USB hub. Most Wi-Fi adaptors get installed easily. *Assuming that's the case for you too*, go to the Raspberry desktop and double-click 'WiFi Config' icon (shown in Fig. 4).

(For connecting to a mobile hotspot like cell phones, the 'WPA2' psk type authentication is used. Select the same from the pulldown menu.)

On a terminal window, type `'iwconfig'` and look for `'wlan0'`. It will show the wireless SSID, it is connected to, and the frequency, signal level, noise level and all other information—above all, it shows that it is connected. Type `'ifconfig'` to see the IP address it has been assigned by the wireless station (see Fig. 6).

Type 'ping www.raspberrypi.org.'

Things may go awry at times and you may find that your Wi-Fi adaptor is not working. First, check the list of workable USB Wi-Fi adaptors. If your Wi-Fi adaptor is not in the list, better buy another one. Alternatively, here is an indirect way to tweak it. Enter `'dmesg | grep usb'` command on a terminal prompt.

This command will furnish details of the kernel ring buffer of all the devices installed starting with the word 'usb'. Look for '802.11' and the manufacturer's name like 'realtek' and 'ralink' associated with it. It means the hardware is built on that chip and the Raspberry Pi needs a driver for it.

Now use your Internet-enabled computer and issue the following command:

The command will give you details of the packages available for that manufacturer. Go to the manufacturer's site and download the Debian wireless package (firmware-realtek.deb, firmware-ralink.deb, etc) from the website. Copy this package in the '/home/pi/' folder of the SD card and issue the following command at the Raspberry Pi:

This will install the right firmware on the Raspberry Pi. Once the firmware is installed, disconnect the wireless adaptor and reconnect it to the Raspberry Pi. This will restart the kernel search process for the right software driver. Next time when the Raspberry Pi is started and the wireless adaptor is connected, it will automatically load the right driver.

With the firmware installed, setting up the connection is pretty straightforward now. Go back to the 'desktop,' double-click 'wpa_gui' icon and then follow the steps explained already for wireless LAN set-up. ●

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